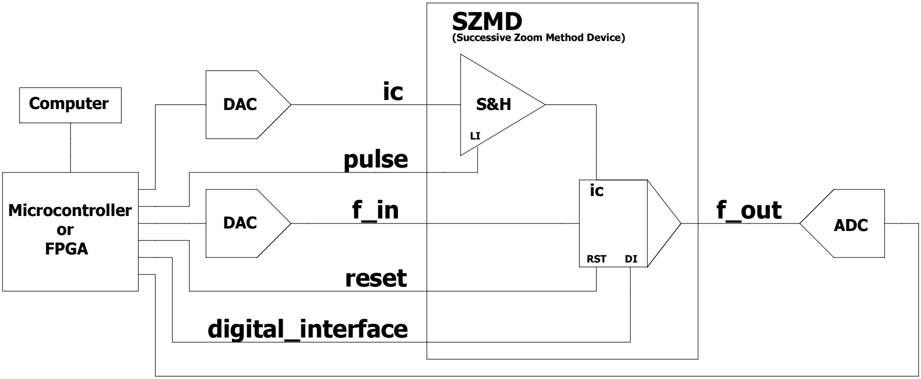


Figure 4 – **Example of an INTEGRATOR SZMD.** A computer — with or without the aid of GPGPUs — commands hardware implemented in a microcontroller or FPGA, which in turn physically configures the integrator block through the digital interface (“digital_interface”). The integrator can be initialized via a *reset* pin, discharging the capacitor. A digital-to-analog converter (DAC) digitally receives the value of the initial conditions and converts it to an analog value. The microcontroller or FPGA sends a pulse so that the *Sample & Hold* block samples and holds this value, which is applied to the “IC” pin of the INTEGRATOR block, thus defining the initial conditions of the analog integration. A second DAC receives digital data of the function to be integrated (“*f_in*”) and converts it into the integrator’s analog input signal. The integrated signal (“f_out”) is captured by an analog-to-digital converter (ADC), reconverting the integrated value into a sequence of digital values, which return to the microcontroller/FPGA and, subsequently, to the computer. Several of the lines in the diagram represent buses.